Trainings

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Disconnect the batteries to prevent accidental engine starting. See equipment manufacturer service information.
- Remove the transmission, clutch, and all related components. See equipment manufacturer service information.

Remove

⚠ WARNING ⚠

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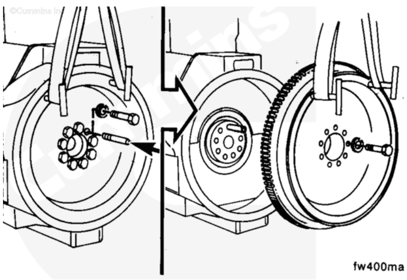


Use two 10 - 22 x 1.5 mm guide studs to prevent the flywheel from rotating. Remove two capscrews and install the guide studs.

Use a hoist, two T-handles, and a lifting sling. Install the T-handles.

Remove the remaining capscrews.

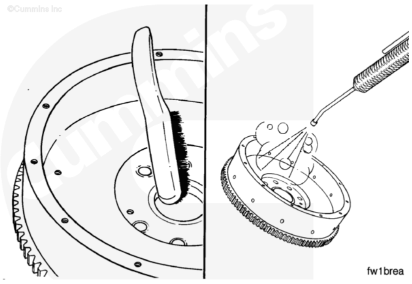
Remove the flywheel.

Use a mallet to tap the flywheel from the crankshaft, if necessary.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

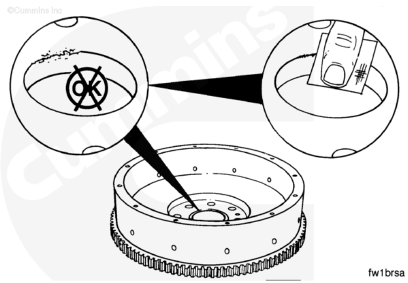
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use a wire brush to clean the crankshaft pilot bore.

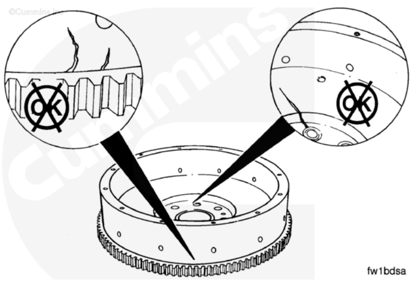
Use steam or solvent to clean the flywheel and capscrews.

Dry with compressed air.

Inspect the flywheel pilot bore for nicks or burrs. Use a crocus cloth to remove small nicks and burrs.

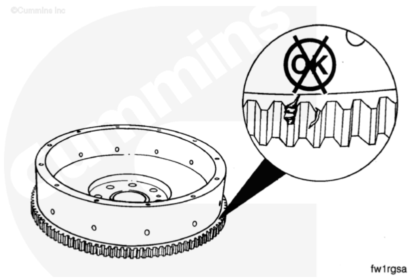


Inspect the flywheel for cracks.



Inspect the flywheel ring gear teeth for cracks and chips.

If the ring gear teeth are cracked or broken, the ring gear **must** be replaced.



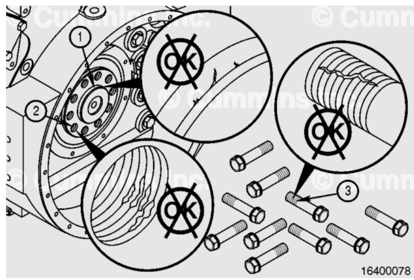
Inspect the crankshaft pilot (1) and the ten threaded capscrew holes (2) for burrs, thread damage, or other damage.

If damage to the threaded holes or the pilot is found, the crankshaft may need to be replaced.

Inspect the ten capscrews (3) for thread damage or burrs.

If damage to the capscrew threads is found, the capscrews **must** be replaced.

Note : Count the number of punch-marks on the head of the capscrew. If five marks are present, the capscrew must be replaced.



Install

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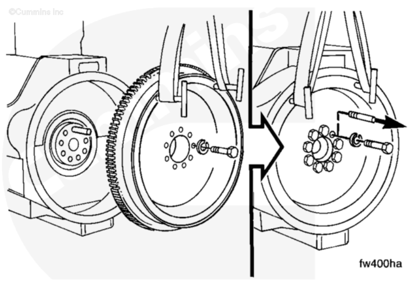


⚠ CAUTION ⚠

The flywheel mounting capscrews must be a minimum of Society of Automotive Engineers (SAE) Grade 8 with rolled threads. The flywheel mounting washers are hardened plain washers. Use identical replacement parts or damage to the flywheel and drivetrain can occur.

Apply a light film of engine oil, SAE 15W40, to bolt threads and bolt underhead during installation.

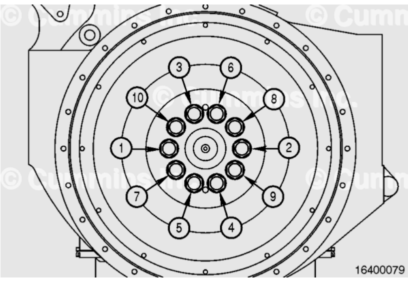
Allow the excess oil to drip off the capscrews.

Do **not** lubricate the threads in the crankshaft.

A guide stud will help during assembly. Install the flywheel, washers, and capscrews. The flywheel **must** be firmly against the crankshaft.

⚠ CAUTION ⚠

If using impact wrench or DC torque tool, do not allow torque tool to exceed 16 rpm. If the torque tool spindle speed isn't able to be adjusted, hand start and tighten each capscrew manually to avoid galling. Manually break at least three threads before loosening.



Use the following steps and the torque sequence shown to install the flywheel.

Torque Value:

1. 200 n•m [148 ft-lb]
2. 460 n•m [339 ft-lb]
3. Tighten 90 degrees

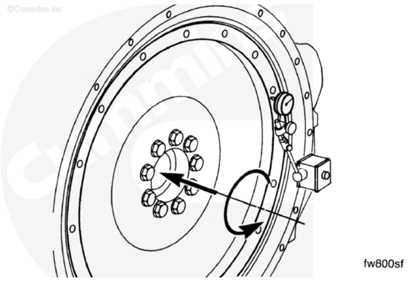
Mark the head of each capscrew with a punch-mark each time it is completely tightened.

Note : Capscrews must not be reused after five punch marks are present on the capcscrew head.

Measure

Measure the face runout of the flywheel.

Note : The crankshaft end clearance must be pushed or pulled in the same direction each time a point is measured.



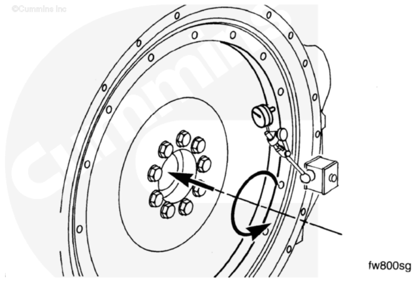
Attach an indicator as shown. Measure the flywheel alignment at four equally spaced points.

Measure the distance from the center of the crankshaft to the indicator tip. Multiply the distance by 0.001 to obtain the maximum face runout.

If the flywheel face runout is **not** within specifications, check for interference between the flywheel and the crankshaft.

Measure the radial runout of the flywheel.

Attach an indicator as shown. Continue to look at the indicator while rotating the engine.



Flywheel Radial Runout		
mm		in
0.13	MAX	0.005

If the radial runout is **not** within specification, the pilot on the flywheel is **not** positioned correctly on the crankshaft.

If the pilot is damaged, the flywheel **must** be replaced.

Finishing Steps

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⚠ WARNING ⚠

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- Install the transmission, clutch, and all related components. See equipment manufacturer service information.
- Connect the batteries. See equipment manufacturer service information.

Last Modified: 15-Oct-2018